

[ARCHIVED CATALOGUE]

Electrical Engineering (Computer Networks) (MS)

Under the computer networks option students must satisfy the MS, Electrical Engineering requirements with the exception that only 15 units of EE are required.

Fundamental Courses

It is expected that each student in this program will take or have taken the equivalent of three of the four following fundamental courses:

- [CSCI 402 Operating Systems](#) Units: 4
- [EE 450 Introduction to Computer Networks](#) Units: 4
- [EE 457 Computer Systems Organization](#) Units: 4
- [EE 503 Probability for Electrical and Computer Engineers](#) Units: 4

Note:

With the exception of [EE 503](#), the fundamental courses may also be satisfied by having passed EE placement exams.

Three of the following courses are required:

- [CSCI 551 Computer Networking](#) Units: 4
- [EE 550 Data Networks: Design and Analysis](#) Units: 4
- [EE 555 Broadband Network Architectures](#) Units: 3
- [EE 597 Wireless Networks](#) Units: 4

Note:

If a fourth required course is taken it can be counted toward elective credit.

Suggested elective courses include:

- [CSCI 530 Security Systems](#) Units: 4
- [CSCI 555L Advanced Operating Systems](#) Units: 4
- [CSCI 558L Internetworking and Distributed Systems Laboratory](#) Units: 3
- [CSCI 570 Analysis of Algorithms](#) Units: 4
- [EE 512 Stochastic Processes for Financial Engineering](#) Units: 4
- [EE 532 Wireless Internet and Pervasive Computing](#) Units: 3
- [EE 535 Wireless Communications](#) Units: 4
- [EE 554 Cyber-Physical Systems: A Computing Perspective](#) Units: 4
- [EE 557 Computer Systems Architecture](#) Units: 4
- [EE 558 Optical Fiber Communication Systems](#) Units: 3

- [EE 590 Directed Research](#) Units: 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12
- [EE 649 Stochastic Network Optimization and Adaptive Learning](#) Units: 4
- [EE 650 Advanced Topics in Computer Networks](#) Units: 3
- [EE 659 Advanced Topics in Interconnection Network Design and Analysis](#) Units: 4

Note:

Any other course must be approved by a faculty adviser.

Total units required for the degree: 27
